



New Cellular Automata. Models, Algorithms, Applications and Anticipation Accounting

Aleksander Makarenko^(✉) 

Institute of Applied System Analysis at National Technical University of Ukraine (I. Sykorski KPI), Prospect Pobedy 37, Building 35, Kyiv 03056, Ukraine
makalex51@gmail.com

Abstract. New generalizations of cellular automata CA are described. This survey contains recent results published in research journals, conference proceedings etc. Cellular automata had been considered in the restricted regions of space. In this work, we propose the concepts for modelling logical gates by propagation of the gliders of cellular automata in a bounded domain and consider special roles for collisions of gliders with walls and obstacles. Also, the general formulations and properties of cellular automata with the strong anticipatory property of cells (introduced by D. Dubois) are considered. Multivalued behaviour (hyperincursion) of solutions of such CA is described. This poses new research problems of computation theory related to the presumable multivaluedness of cellular automata with strong anticipation properties. Extending of classical automata, Turing machine and algorithms had been proposed. Also, some relations of such cellular automata and quantum mechanics are discussed. Applications of cellular automata can remind well-known processes in real life like football games; migration in science and higher education; epidemic spreading; and artificial life.

Keywords: Cellular automata · new models · applications · anticipation

1 Introduction

Cellular automata (CAs) are a very interesting class of models, both in terms of orthographic and applications (see [1–6] and many others). The idea of classical cellular automata is roughly as follows:

- the space is divided into identical elements (for example, as in the classic game “life” - into square cells on the plane);
- each element has a finite number of states (in the game of life, there are only two states);
- classical CA models are considered at discrete moments of time;
- dynamic laws for changing the states of the cell from the previous moment to the next moment are formulated;
- dynamic laws for changing the states of each cell are constructed in such a way that their formulas depend only on the values of cell states in the local neighbourhood of the cell (for example, in the game “life”, a neighbourhood of 4 cells (Neumann neighbourhood), or a neighbourhood of 8 cells (Moore neighbourhood)).

All these settings hold for classical cellular automata which have a large number of both theoretical studies and applied applications. However, non-classical CAs are also being developed in parallel. For example, CAs with fuzzy states of elements, CAs with memory, CAs with special geometry of the underlying space, including those on graphs, probabilistic CAs, and many others.

The proposed article describes the results of constructing generalized models of cellular automata, studying the properties of such CAs and their applications. The following are mentioned among such CAs: Non-classical cellular automata; Elements constructed from CA for emulation of logical operation; Non-determinism in CA; Demography and migration; Crowd movement; Football modelling; Epidemic modelling.

Particularly interesting are cellular automata with anticipation (accounting of the future), which have the possibility of multiple states of cellular automata (i.e., many states of one cell element at a fixed time). The article concludes with several directions for further research.

2 Short Description on Investigations on CA

2.1 Some Theoretical Investigations on CA

Here we pose the short titles and references on some of our theoretical research related to the cellular automata: non-regular elements in CA [7]; probability in local withinities [8]; multilevels CA [9]; modeling of logical gates by CA in restricted domain [10]; CA with the anticipation property [11]; artificial Life in cellular space [12]; CA of game “Life” in restricted domain [13]; identification problems for CA rules [14].

2.2 Some New Applications of Cellular Automata

Here we pose the short titles and references on some of our research related to the cellular automata applications: prediction of electoral processes with cellular automata [15]; prediction of pedestrian crowds movement [16]; football modeling by cellular automata [17]; epidemic modeling [18]; demography modeling [19]; mobility of students and researches [20].

2.3 List of Multivalued Cellular Automata Investigations

Here we pose the short titles and references on some our theoretical research related to the cellular automata in multivalued case, especially of CA with strong anticipation: cellular automata with strong anticipation [21]; multivalued cellular automata [22]; CA and non-determinism [23]; CA in quantum mechanic [24]; CA and theory of computation [25]; complex-valued CA with branching [26].

3 The Three Most Interesting Research Results

3.1 Multivalued Cellular Automata with Strong Anticipation

In this paragraph, we give some results on multivalued CA with strong anticipation research. Considered in literature until now cellular automata (CA) had dynamic laws in which the state on the next temporal step (with discrete time) relied only on the state of elements in the local neighbouring in the previous moments of time (see for example [1–3]). Consideration of the possibility of foresight in the dynamics of elements leads to equations with anticipation.

The term ‘anticipation’ has been introduced in active use in biology and applied mathematics by Robert Rosen [27]. So this name is well known for common researchers. Less known is the contribution by Daniel Dubois to strong anticipation. Since the beginning of 90-th in the works by D.Dubois – see [28, 29] the idea of strong anticipation had been introduced: “Definition of an incursive discrete strong anticipatory system ...: an incursive discrete system is a system which computes its current state at the time, as a function of its states at past times , ..., $t - 3$, $t - 2$, $t - 1$ present time, t , and even its states at future times $t + 1$, $t + 2$, $t + 3$, ...

$$x(t + 1) = A(..., x(t - 2), x(t - 1), x(t), x(t + 1), x(t + 2), ..., p) \quad (1)$$

where the variable x at future times $t + 1$, $t + 2$, $t + 3$, ... is computed in using the equation itself”.

In the general case it follows formal dynamic equations (for the case of discrete-time of $t = 0, 1, ..., n, ...$) (see for example [21–24, 28, 29]):

$$s_i(t + 1) = G_i(\{s_i(t)\}, ..., \{s_i(t + g(i))\}, R), \quad (2)$$

where the R is the set of external parameters (environmental, control etc.) is, $\{s(t)\}$ - the state of the system in the moment of t , $g(i)$ - horizon of foresight, $\{G\}$ – a collection of nonlinear functions for the calculation of change of the state of elements. These equations are poorly explored even in mathematics. Therefore here we will pose only the pair of the simplest examples necessary for the exposition the ideas of the article.

One of the simplest examples of cellular automata is the well-known game «Life» by J.Conway ([1–3]). In the «classic» game “Life» by Conway every cell can have only two states (we will designate them 0 and 1, $S = \{0, 1\}$); neighbouring, which is used for the record of local rules of change –Moor’s neighbouring, consisting of eight neighbouring cells:

In most descriptions of the game «Life» following biological associations are proposed: two states of the cell 0 or 1 are interpreted as «dead» or «living» cells. Then the changes of the state of cells can be described by the following rule:

Let in the moment of t some subset of cells is made by living cells. In the moment of time of $t + 1$ those cells which in the moment of t fulfilled conditions will be living:

1. If a living cell has two or three living neighbors in its neighboring, it remains living in the next moment of time. Otherwise, the cell dies.
2. If a dead cell has exactly three living neighbors in neighboring, then she becomes «living» in a next moment of time.

All «births» and «deaths» take place simultaneously.

Rules for calculation of the states in the next moment of time in the game «Life» can be written down how some functions of transition are, the representing states in the local neighboring of cell in the state of the cell in the next moment of time. At the consideration of foresight (in the simplest variant) this state relies also on the possible states of cells in the local neighboring also and in the next moment of time.

These results were described in a separate publication (see for example [20–24])), but one of the most fundamental results of accounting anticipation which is important for us is the possibility of the existence of many branches of solutions (possibility of a few states of every cell in a cellular automaton in a given moment of time). Here in the picture, we pose one of the examples of such solutions (Fig. 1):

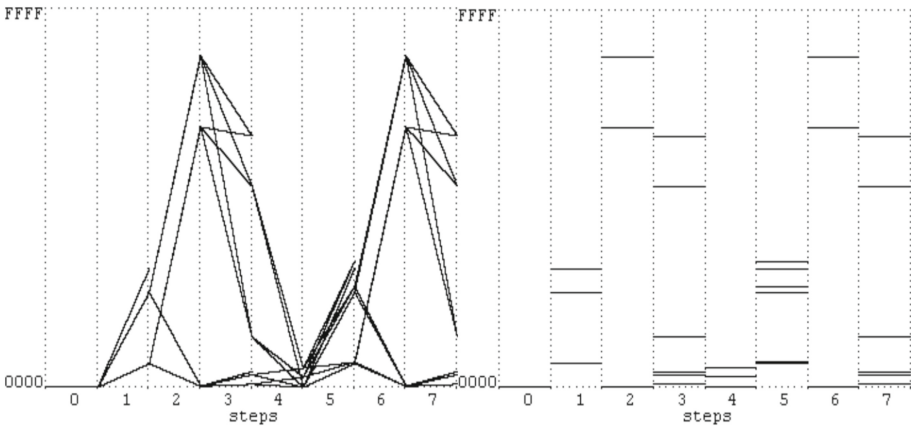


Fig. 1. Solution of CA with strong anticipation with a few possible states (on abscise axis the steps of discrete time are postponed, and on a y-axis the number of different system configurations is represented (each configuration is the collection of the single-valued states of all cells of the system)).

In the given example we also display the number of the possible states. Further possibilities, as well as details, have been suggested in other publications. So, the basic distinctive feature is the multiplicity of solutions and presumable configurations at every instant. Possible consequences of these facts and the problems of reconstruction of automata theory may be considered.

3.2 Modelling of Logical Gates by CA in a Restricted Domain

As far as we are concerned, the implementation of logic gates we have made is based on three points. First of all the information is no longer represented by a glider beam as described above but by a single glider. We no longer need to resort to glider guns, which is a big novelty. Then we set up a particular network with “walls”. Each logical gate is a particular configuration of space, a feature that is exploited by bouncing the gliders wisely as described in part 4. Finally, the method relies on collisions between gliders. And especially the two collisions that we analyzed in part 2.

Generally, the logic gates set up have two ducts at the top (representing the two inputs). A glider in the conduit means that the entry is at 1 otherwise it is at 0 (see Fig. 2). And a conduit down (representing the exit). The particular configuration of the rest of the network will allow or not to obtain a glider in the lower conduit depending on the nature of the logic gate.

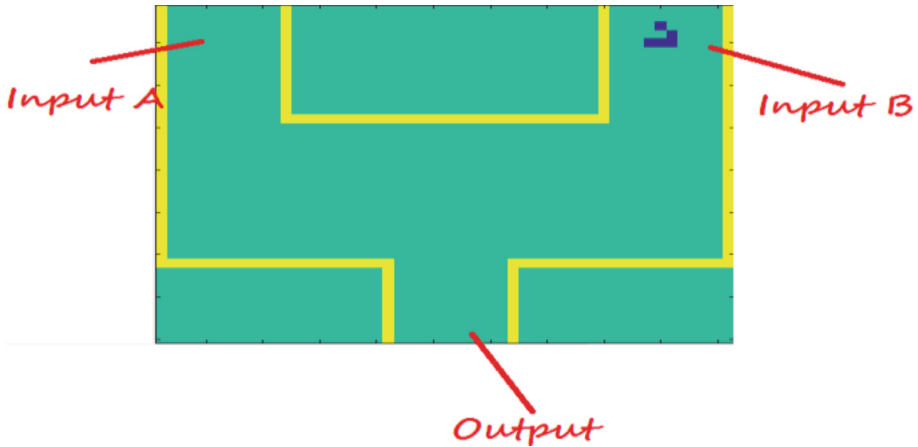


Fig. 2. Logic gate with two inputs $A = 0$ and $B = 1$.

Each of the four logic gates has been implemented. Namely: the NOT gate, the OR gate, the AND gate and the XOR gate.

3.3 Mobility of Students and Researchers

The problem of migration of researchers and students is very relevant, especially for Ukraine in the current situation. A method of forecasting this process is proposed. Computer programs for cellular automata models are proposed and implemented. A flexible structure of computer programs is developed and the structure of software modules and their interaction is proposed. The migration calculation was made for a server-based solution based on Guscuit for an unlimited number of users with the ability to access the application from any mobile device. The implementation used the Model-View-Controller concept. The model provides data and responds to controller commands. The view is responsible for displaying the model data. This technology allows you to modify each component independently of each other (Fig. 3).



Fig. 3. Final distribution of Ukrainian students in Western Europe (one of the scenarios).

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